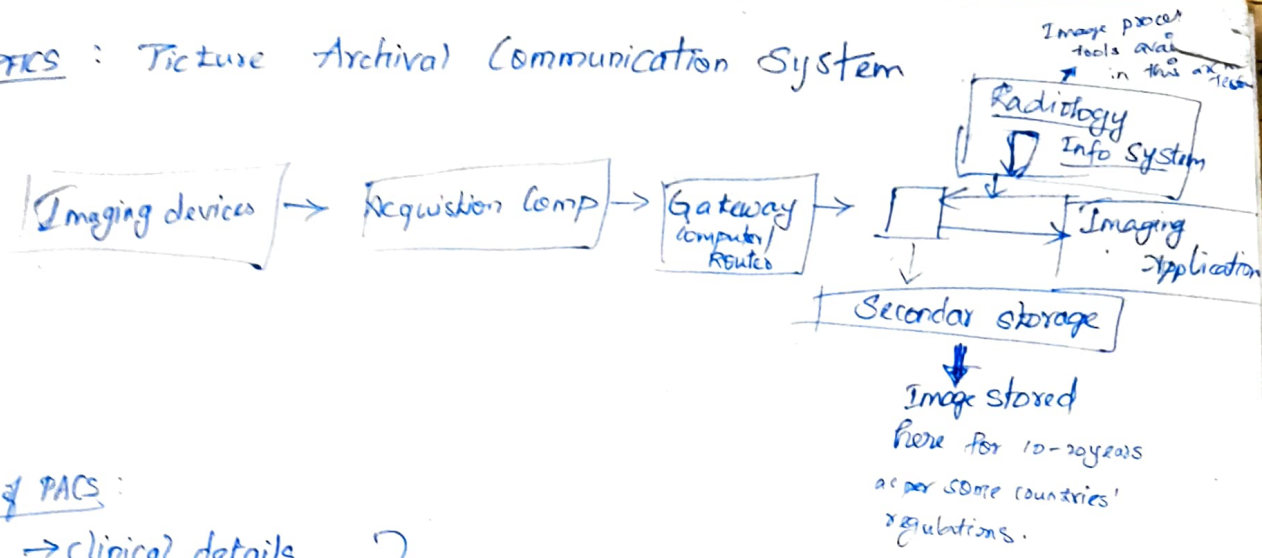
## PACS : Picture Archival Communication System




**Role of PACS:**

→ clinical details
→ Radiological report } Stored with the digital image [Meta data]

CT - 40 scans/day
MRI - 22 Scans for calibrated system } operational dependant.

⇒ Acquistion (CT, MRI, x-ray, USG) → PAC server → Web

PAC server → Report room
Seperate from acquistion room.
→ Some countries like US, where voice over report added to the data

**Subsystems of PACS:** 5 - sub systems.

**HIS** → Hospital Info System, gives basic ~~info~~ struct about patient info
Unique Patient ID is used
HIMS is used [Hosp. Info. Management System]: manages all entry.

**RIS:** Radiology Info System. Patient info collected from HIS.

If a patient needs MRI scan, HIS gives what to scan.

HIS ⇄ RIS
PAC ⇄ RIS

→ Bidirectional integration
→ input of demographic data only once
→

## DDR :-

Xray → CSI $\xrightarrow{\text{light}}$ Amorphous silicon Panel $\xrightarrow{\text{electrons}}$ Read out Electronics → Digital Signal

Xray → Selenium (Amorphous silicon panel) → Readout Electronics

## DBMS:

LTA ← Short term server → Web
Short term server ⇄ work station
Image acq → Short term server

⇒ SSD used for very fast memory.
⇒ LTA ⇒ Long Term storage, Magnetic Tapes used.
⇒ Designing PAC system require three structures
Short Term, Medium Term, Long Term.
This decides architecture and cost of system.

Image Server

Short Term: RAID config is required.

⇒ Redundancy of data is decided [data stored 2 times. 2x redundancy. data retrieved from other drive if one fails].

⇒ Distributed copy of data in 5 or 6 different places.

Pacs Work stations:

Display Work station :

Review work station :